CV

Erol Orel
Current position(s): Research associate / Scientific collaborator, Lecturer / Reader
Academic age: 3 year(s)

Education

Degree	Organisation	Duration
PhD / Dr.: PhD in Biomedical Sciences Prof. Olivia Keiser	Université de Genève - GE, CH Faculty of Medicine	09.2019 - 12.2023 4 year(s) 4 month(s)
Further Advanced Studies: I-Teams Medical Innovation Program	Université de Genève - GE, CH Faculty of Medicine	01.2023 - 10.2023 10 month(s)
Further Advanced Studies: Financial Risk Manager (FRM)	Global Association of Risk Professionals, US Professional certification programme, FRM Part I and Part II	02.2018 - 08.2018 7 month(s)
Further Advanced Studies: D.E.A. in Econometrics	Université de Genève - GE, CH Faculty of Economics	09.2004 - 06.2006 1 year(s) 10 month(s)
Master: Quantitative Economics N/A	Université de Genève - GE, CH Faculty of Economics	09.2000 - 06.2004 3 year(s) 10 month(s)

Employment

Role	Organisation	Duration
Lecturer / Reader	Université de Genève - GE, CH	09.2024 - 08.2027 3 year(s)

Role	Organisation	Duration
	Faculty of Medicine, MAS en santé publique	
Research associate / Scientific collaborator Prof. Olivia Keiser	Université de Genève - GE, CH Faculty of Medicine, Institute of Global Health	07.2019 - 08.2027 8 year(s) 2 month(s)
Group leader	Market Intuitive Investment Managers, CH Risk Management	06.2014 - 05.2018 4 year(s)
Financial Risk Manager / Senior Quantitative Analyst	Steenman Asset Management, CH Asset Management Services	06.2007 - 02.2014 6 year(s) 9 month(s)
Senior researcher	Cantonal Service of Statistical Research and Information, CH SCRIS	01.2007 - 05.2007 5 month(s)
Research associate / Scientific collaborator Prof. Gabrielle Antille-Gaillard	Université de Genève - GE, CH Laboratory of Applied Economics (LEA), project Economic Indicators of the Geneva Economy, LEA-PICTET-OCSTAT	09.2005 - 08.2006 1 year(s)
Teaching Assistant - Statistical Methods / Statistical Modeling	Université de Genève - GE, CH Faculty of Economics	09.2005 - 08.2006 1 year(s)

Major achievements

Achievement 1

During my doctorate I used machine learning on large population surveys to predict individual HIV status from social and behavioural characteristics across East and Southern Africa, and I examined where a model learnt in one country held in another [1]. Related work in the group addressed treatment interruption in South Africa [2] and country clusters by behaviour [3]. During the pandemic I applied the same skills to Swiss and African data, comparing the clinical severity of Delta and Omicron in a Swiss university hospital [4], contributing to the Swiss age-structured scenario model used for planning under time pressure [5], and mentoring the analysis of a seroprevalence study in Cameroon as senior

statistician [6]. As last author I supervised the statistical analysis and the code of a machine-learning classification of cardiovascular disease across WHO STEPS surveys, accepted in BMC Global and Public Health [7]. These studies gave me direct experience of when a model transfers between settings, the question at the centre of my Ambizione project.

[1] journal-article. Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. “Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa.” PLoS ONE (2022); 17(3): e0264429. DOI. Open Access.
[2] journal-article. Esra R, Carstens J, Le Roux S, Mabuto T, Eisenstein M, Keiser O, Orel E, Merzouki A, De Voux L, Maskew M, Sharpey-Schafer K. “Validation and Improvement of a Machine Learning Model to Predict Interruptions in Antiretroviral Treatment in South Africa.” J Acquir Immune Defic Syndr (2023); 92(1):42-49. DOI. Open Access.
[3] journal-article. Merzouki A, Estill J, Orel E, Tal K, Keiser O. “Clusters of sub-Saharan African countries based on sociobehavioural characteristics and associated HIV incidence.” PeerJ (2021); 9: e10660. DOI. Open Access.
[4] journal-article. Orel E, Sobel J, Zanella MC, Iten A, Kaiser L, Keiser O, Vetter P. “Difference in clinical severity between Delta variant and Omicron sub-lineages in a Swiss tertiary center.” CMI communications (2024). DOI. Open Access.
[5] journal-article. Estill, J., Venkova-Marchevska, P., Roelens, M., Orel, E., Temerev, A., Flahault, A., & Keiser, O. (2021). Future scenarios for the SARS-CoV-2 epidemic in Switzerland: an age-structured model. F1000Research, 9, 646. https://doi.org/10.12688/f1000research.24497.2. DOI.
[6] journal-article. Nwosu K, Fokam J, Wanda F, Mama L, Orel E, Ray N, Meke J, et al. “SARS-CoV-2 antibody seroprevalence and associated risk factors in an urban district in Cameroon.” Nature Communications (2021); 12(1): 5851. DOI. Open Access.
[7] preprint. Ng'ambi, W., Estill, J., Keiser, O., Merzouki, F. A., & Orel, E. (2026). Machine learning-based classification of self-reported cardiovascular disease history in Africa using harmonised multi-country WHO STEPS surveys: 2014–2019. medRxiv. https://doi.org/10.64898/2026.02.23.26346870 (accepted for publication in BMC Global and Public Health). DOI.

Achievement 2

I conceived and built LiteRev with Aziza Merzouki and Olivia Keiser, a platform that searches the open literature from plain-language or Boolean queries, removes duplicates, maps the retrieved studies into topics and supports double screening with PRISMA accounting. In its published evaluation it saved 56% of the screening work compared with a manual review (work saved over sampling) [1]. It grew out of a semi-automated review of HIV in Malawi [2]. The tool was funded by CHF 70,000 obtained with the group (UNIGE, Fondation Ernst et Lucie Schmidheiny, Venture Kick, Mimosa) and is used by other teams, including a systematic review of artificial intelligence in emergency medical services within the Franco-Swiss GESICA consortium, which I co-authored [3]. Since 2024 I have extended it into LiteRev-Evidence, a prototype that extracts numbers from studies with their uncertainty, pools them into parameter distributions and fits first forecasting models. These parts exist as a prototype and have not yet been validated as one pipeline. Testing whether such extracted evidence can improve forecasts at the start of a health crisis is the aim of my Ambizione project.

[1] journal-article. Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. “An Automated Literature Review Tool (LiteRev) for Streamlining and Accelerating Research Using Natural Language Processing and Machine Learning: Descriptive Performance Evaluation Study.” J Med Internet Res (2023); 25: e39736. DOI. Open Access.
[2] journal-article. Thiabaud, A., Triulzi, I., Orel, E., Tal, K., & Keiser, O. (2020). Social, behavioural, and cultural factors of HIV in Malawi: a semi-automated systematic review (Preprint). Journal of Medical Internet Research. https://doi.org/10.2196/18747. DOI.
[3] preprint. Edjinedja, K. L., Larribau, R., Orel, E., Cossus, J., Elfahim, O., Yendouname, K., Vaussenat, F., Teodoro, D., Robert-Nicoud, S., Barakat, O., & Desmettre, T. (2026). Artificial intelligence in emergency medical services for disasters and health emergencies: A systematic review [Manuscript submitted for publication]. Machine Learning and Knowledge Extraction.

Achievement 3

Since 2024 I have worked on Geneva's surveillance and emergency data. Within GESICA I mapped the variables and data sources for influenza, COVID-19, RSV and heat waves, in an inventory of 28 sources documented for coverage, resolution, latency, access and quality, and I contribute to the European epidemic-intelligence toolbox of GeoAI4EI (Horizon Europe). I wrote the statistical analysis plan of a cantonal study that links confirmed legionellosis cases to the hot-water installations of buildings (CHF 30,000 each from Services Industriels de Genève and the Office cantonal de l'énergie, a further CHF 30,000 requested from the médecin cantonal), and I carry out its full analysis under ethics approval BASEC 2026-00324. During the pandemic I helped building the COVID-19 epidemiological dashboard of the WHO African Region. Since 2024 I lecture in statistics and epidemiology on the MAS in Public Health, I have supervised eight MAS theses and mentored about six junior researchers at the Institute, and I review occasionally for journals.